

(a) Parametric Flight Sweep & Leak-Free Partitioning

PX4/Gazebo Flight Sweep

4 Yaw Rates (7.5°-90°/s)
× 4 Geometries (C, B, S, H)
48 Executions in Sim

Physical
Gates

Quality Envelope Filter

Excludes 6 Unviable Flights
42 Eligible Flights
31,195 Frames (34.6% Fail)

Seed 42
Disjoint

Flight-Level Quarantine

Train: 15 flights (11,178 frames)
Val: 13 flights (9,606 frames)
Test: 14 flights (10,411 frames)
Train n Test = $\emptyset$ (Zero Leak)

Sensor Stream

Scaler Fit (Train)

(b) Causal Feature Construction & Lightweight Neural Predictor

Flight Telemetry (30 Hz)

- Yaw Rate: ω_z (°/s)
- Optical Flow: $\vec{v}_{\text{flow}}$ (px)
- Keyframe Flag: $r \in \{0,1\}$

N=6 FIFO

15-Dim Lagged Vector

Lags {0, 1, 2, 3, 5}
Span = 165 ms
 $x_t = [\omega_{\text{lag}}, \vec{v}_{\text{lag}}, r_{\text{lag}}]$

Scale

SmallMLP Predictor

15 → 32 → 16 → 1
1,057 Parameters
Latency: 0.128 ms CPU

Sigmoid

Decision Logic

$P(\text{fail}_{\{t+5\}} | x_t)$
K=5 Forward Horizon
 $\theta = 0.50$ Early Warning

Batch Offline

Causal FIFO

Live ROS 2

(c) Three-Tier Empirical Verification Framework

Tier 1: Offline Generalization

14 Disjoint Held-Out Flights
AUROC = 0.8046 | AUPRC = 0.7260
F1 = 0.6484 (Prec 0.581, Rec 0.734)
Zero Optimization Leakage

Tier 2: Causal Sequential Replay

10,481 Frames Replayed Sequentially
79.3% True Early Warnings (522/658)
Median Lead Time: 0.165 s (5 Frames)
Algorithmic Latency: 0.128 ms

Tier 3: Live ROS 2 Deployment

Live ROS 2 Node (rclpy DDS)
192 Hz Sustained Rate (>6× Headroom)
Mean Latency: 0.634 ms (P99 3.71 ms)
93.4% Live Early Warning Rate